



Gaussian Naïve Bayes

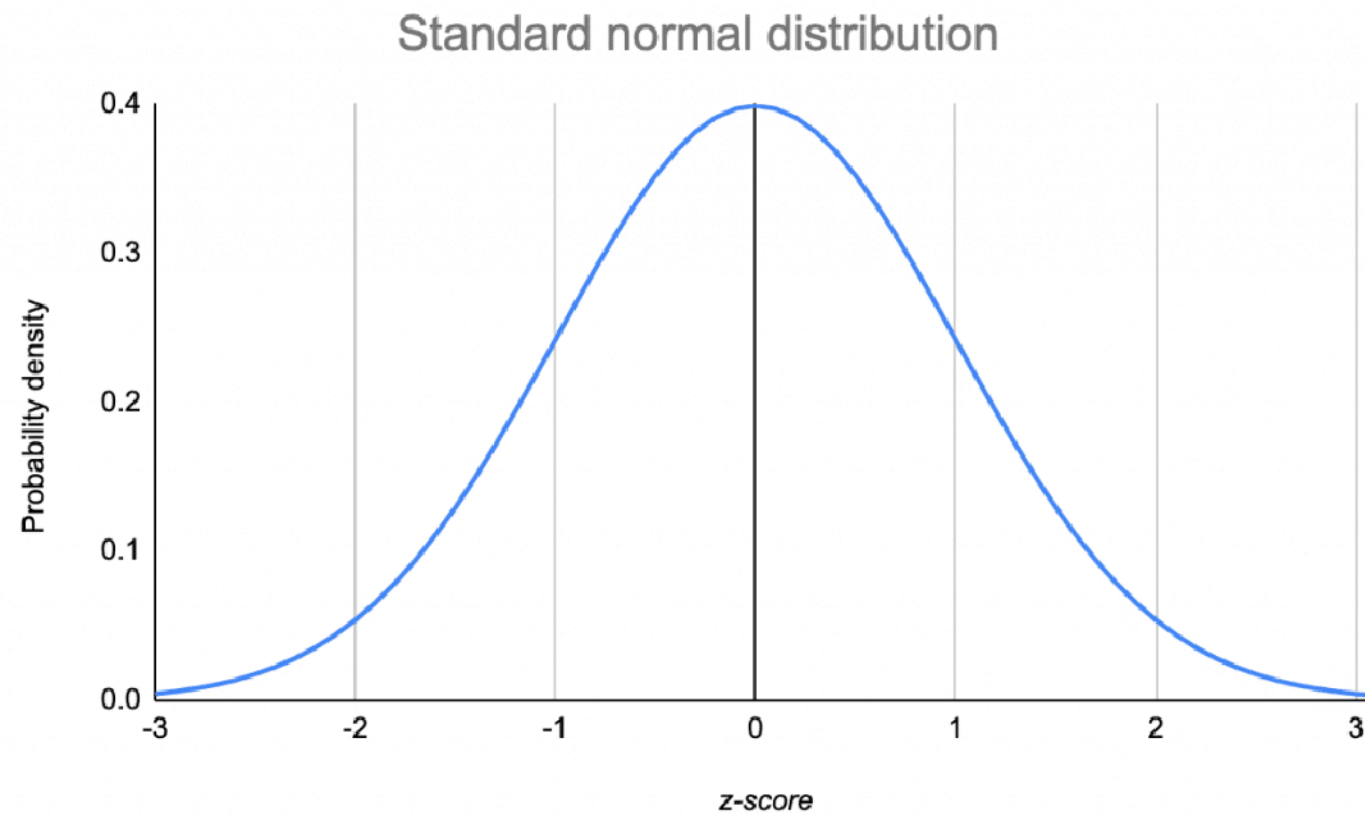
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Gaussian/Normal Distribution

► A lot of things in nature follow the normal distribution, such as people's height, etc.





Gaussian Distribution

► A type of continuous distribution

► (Univariate) probability density function:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

► where μ and σ^2 are the mean and variance of the distribution.



Gaussian Distribution (cont.)

► (Multivariate) probability density function:

$$f_{\mathbf{X}}(x_1, \dots, x_k) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(\mathbf{X} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{X} - \boldsymbol{\mu})}$$

$$\mathbf{X} = (X_1, \dots, X_k)^T$$

► where $\mu_{k \times 1}$ and $\Sigma_{k \times k}$ are the mean vector and covariance matrix of the distribution:

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] = [\mathbb{E}[X_1], \dots, \mathbb{E}[X_k]]^T$$

$$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$



Gaussian naïve Bayes (GNB)

► How to distinguish children from adults, using a Gaussian naïve Bayes classifier?

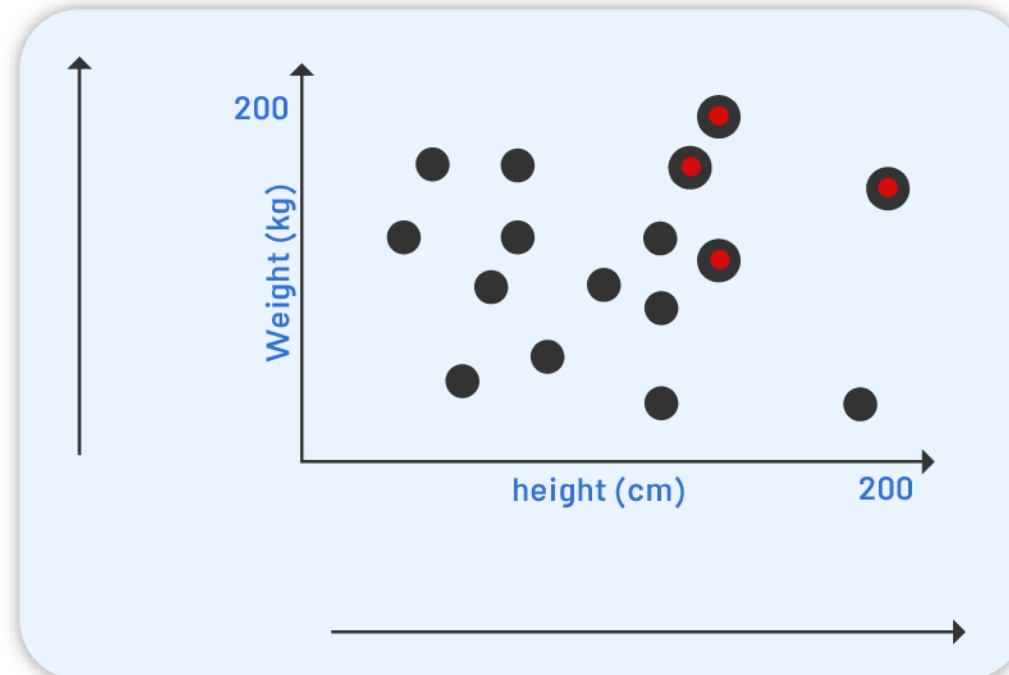
► Classes: {a,c}

► Features: height and weight (continuous inputs)

► Training data $\{h_i, w_i, y_i\}$

► 12 children (black)

► 4 adults (red)



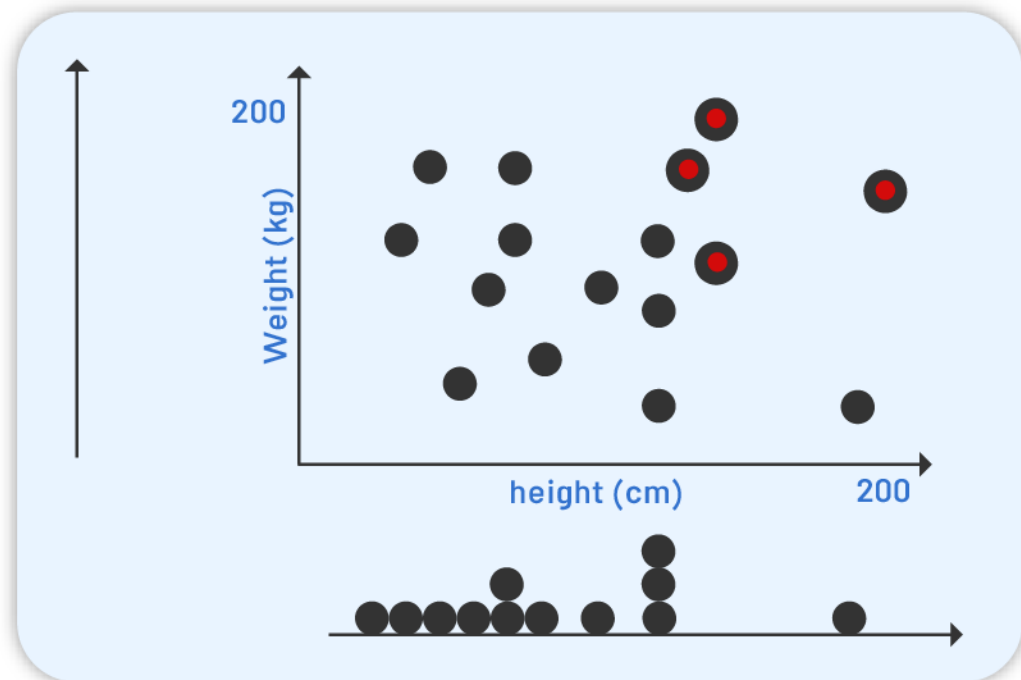


Prior Probabilities

► Prior class probabilities:

► $p(Y = a) = 4/16 = 0.25$

► $p(Y = c) = 12/16 = 0.75$





Children's Features

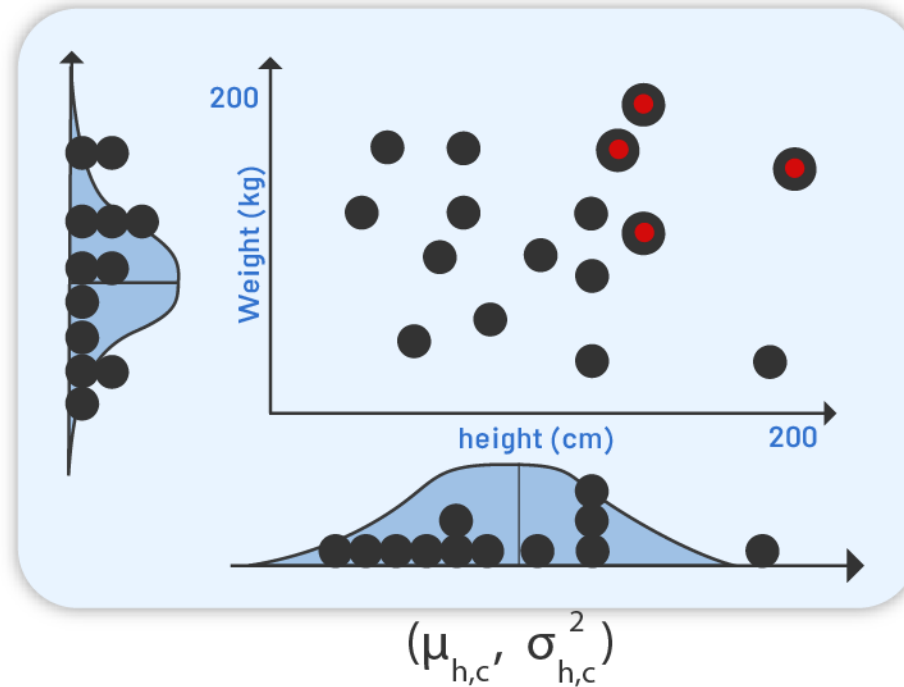
► Assuming that children's heights and weights follow the Gaussian distribution:

$$\mu_{h,c} = \frac{1}{12} \sum_{i:y_i=c} h^i$$

$$\sigma_{h,c}^2 = \frac{1}{12-1} \sum_{i:y_i=c} (h^i - \mu_{h,c})^2$$

$$\mu_{w,c} = \frac{1}{12} \sum_{i:y_i=c} w^i$$

$$\sigma_{w,c}^2 = \frac{1}{12-1} \sum_{i:y_i=c} (w^i - \mu_{w,c})^2$$

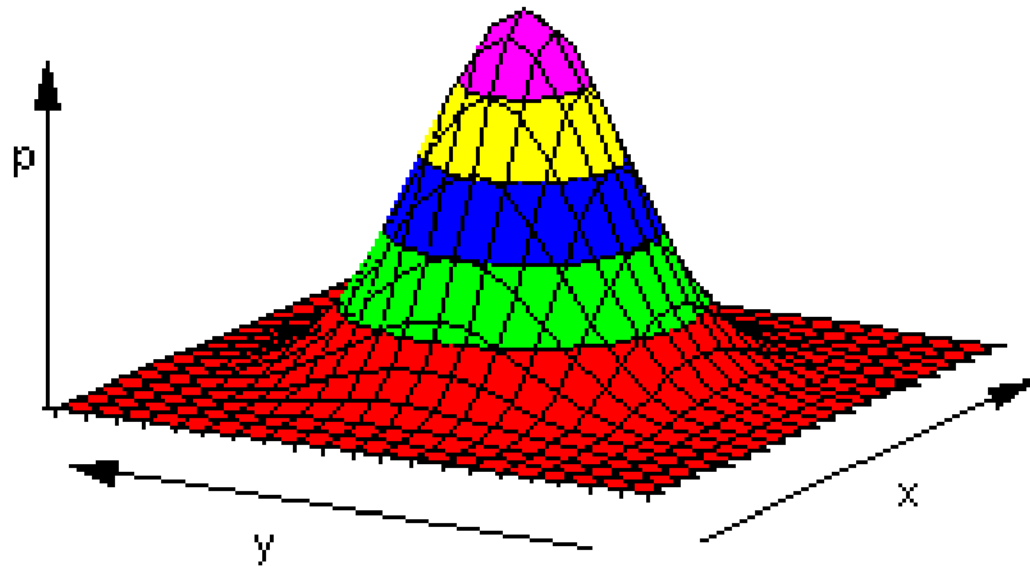




2- Dimensional Gaussian Distribution

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix}$$

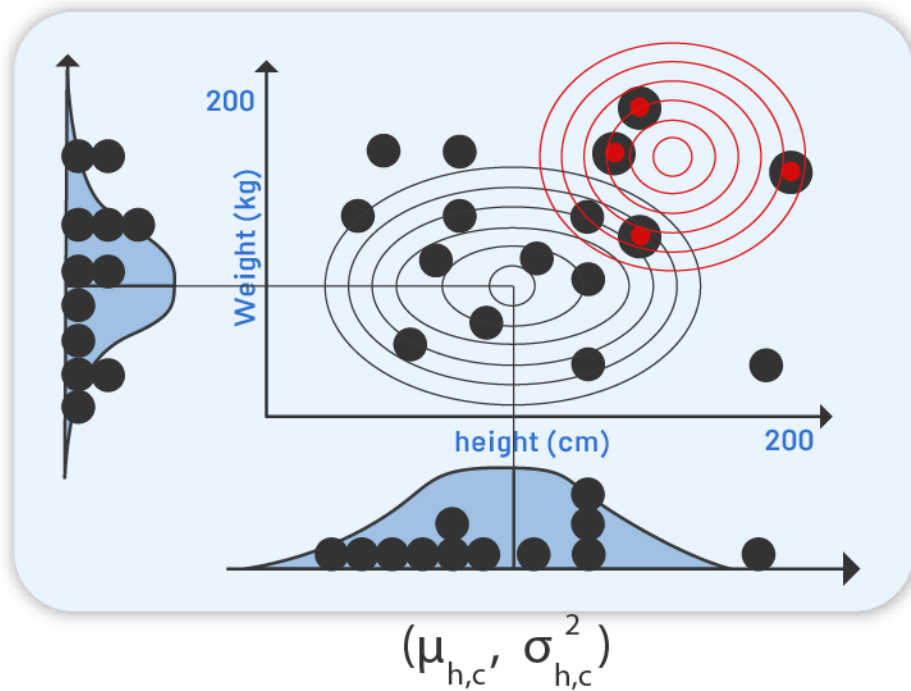
where $\rho = \frac{\text{Cov}(X,Y)}{\sigma_X\sigma_Y}$ is the correlation coefficient between X and Y.





Independence Assumption

► Multiplying two Gaussian distributions



► The same thing for the adult's features



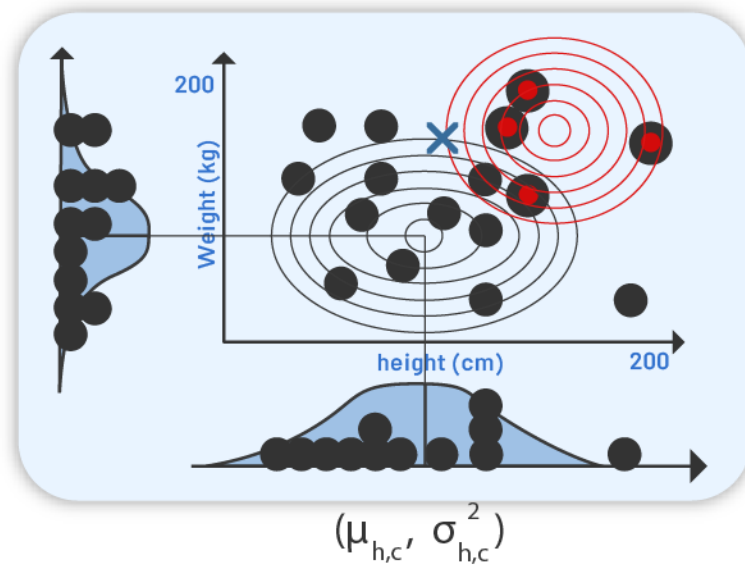
Test Phase

► $X = \langle h_X, w_X \rangle$ (test data)

$$\text{► } p(h_X | Y = a) = \frac{1}{\sqrt{2\pi\sigma_{h,a}^2}} \exp\left(-\frac{(h_X - \mu_{h,a})^2}{2\sigma_{h,a}^2}\right)$$

$$\text{► } p(w_X | Y = a) = \frac{1}{\sqrt{2\pi\sigma_{w,a}^2}} \exp\left(-\frac{(w_X - \mu_{w,a})^2}{2\sigma_{w,a}^2}\right)$$

► Similarly compute $p(h_X | Y = c)$ and $p(w_X | Y = c)$





Classification

► Classification Rule:

$$y_{MLE} = \operatorname{argmax}_{y \in \{a, c\}} p(Y = y | X) = \operatorname{argmax}_{y \in \{a, c\}} p(X | Y = y) p(Y = y)$$

Test data: $X = \langle h_x, w_x \rangle$

$$\text{If } p(\langle h_x, w_x \rangle | Y = a) p(Y = a) > p(\langle h_x, w_x \rangle | Y = c) p(Y = c)$$

Independence assumption



$$\text{If } p(h_x | Y = a) p(w_x | Y = a) p(Y = a) > p(h_x | Y = c) p(w_x | Y = c) p(Y = c)$$

Then Class is adult

Else Class is children



A Gentle Correction

► Population mean vs. Sample mean

$$\mu_{h,c} = \frac{1}{12} \sum_{i:y_i=c} h^i \rightarrow \bar{X}_{h,c} = \frac{1}{12} \sum_{i:y_i=c} h^i$$

► Population variance vs. Sample variance

$$\sigma_{h,c}^2 = \frac{1}{12-1} \sum_{i:y_i=c} (h^i - \mu_{h,c})^2 \rightarrow S_{h,c}^2 = \frac{1}{12-1} \sum_{i:y_i=c} (h^i - \bar{X}_{h,c})^2$$



Parametric Form of $P(Y|X)$ for GNB Classifier

► Assumptions

- Y is boolean, governed by a Bernoulli distribution
- $X = \langle X_1, X_2, \dots, X_n \rangle$, where X_i is a continuous random variable
- For each X_i , $P(X_i|Y = y_k)$ is a Gaussian distribution of the form $N(\mu_{ik}, \sigma_i)$
- For all i and $j \neq i$, X_i and X_j are conditionally independent given Y



GNB Parametric Form of $P(Y|X)$

► We derive the parametric form of $p(Y|X)$ that follows from the set of GNB assumptions:

$$p(Y = 1|X) = \frac{p(X|Y = 1)p(Y = 1)}{p(X)} = \frac{p(X|Y = 1)p(Y = 1)}{p(X|Y = 1)p(Y = 1) + p(X|Y = 0)p(Y = 0)}$$
$$= \frac{1}{1 + \frac{p(X|Y = 0)p(Y = 0)}{p(X|Y = 1)p(Y = 1)}}$$



GNB Parametric Form of $P(Y|X)$

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$$\begin{aligned} p(Y = 1|\mathbf{X}) &= \frac{p(\mathbf{X}|Y = 1)p(Y = 1)}{p(\mathbf{X})} = \frac{p(\mathbf{X}|Y = 1)p(Y = 1)}{p(\mathbf{X}|Y = 1)p(Y = 1) + p(\mathbf{X}|Y = 0)p(Y = 0)} \\ &= \frac{1}{1 + \frac{p(\mathbf{X}|Y=0)p(Y=0)}{p(\mathbf{X}|Y=1)p(Y=1)}} = \frac{1}{1 + \exp\left[\ln \frac{p(\mathbf{X}|Y=0)p(Y=0)}{p(\mathbf{X}|Y=1)p(Y=1)}\right]} \\ &= \frac{1}{1 + \exp\left[\ln \frac{p(\mathbf{X}|Y=0)}{p(\mathbf{X}|Y=1)} + \ln \frac{p(Y=0)}{p(Y=1)}\right]} = \frac{1}{1 + \exp\left[\sum_{i=1}^n \ln \frac{P(x_i|Y=0)}{P(x_i|Y=1)} + \ln \frac{p(Y=0)}{p(Y=1)}\right]} \quad (\text{Eq. I}) \end{aligned}$$



GNB Parametric Form of P(Y|X)

► The sigma term in the denominator:

$$\begin{aligned}\sum_{i=1}^n \ln \frac{p(x_i|Y=0)}{p(x_i|Y=1)} &= \sum_i \ln \frac{\frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(X_i - \mu_{i0})^2}{2\sigma_i^2}\right)}{\frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(X_i - \mu_{i1})^2}{2\sigma_i^2}\right)} \\&= \sum_i \ln \left(\exp \left(\frac{(X_i - \mu_{i1})^2 - (X_i - \mu_{i0})^2}{2\sigma_i^2} \right) \right) = \sum_i \frac{(X_i^2 - 2X_i\mu_{i1} + \mu_{i1}^2) - (X_i^2 - 2X_i\mu_{i0} + \mu_{i0}^2)}{2\sigma_i^2} \\&= \sum_i \frac{2X_i(\mu_{i0} - \mu_{i1}) + \mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2}\end{aligned}$$



GNB Parametric Form of P(Y|X)

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GNB Parametric Form of $P(Y|X)$

$$I, II \Rightarrow p(Y = 1|X) = \frac{1}{1 + \exp[\sum_{i=1}^n \left(\frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2} X_i + \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2} \right) + \ln \frac{p(Y = 0)}{p(Y = 1)}]}$$



GNB Parametric Form of $P(Y|X)$

$$I, II \Rightarrow p(Y = 1|X) = \frac{1}{1 + \exp\left[\sum_{i=1}^n \left(\frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2} X_i + \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2}\right) + \ln \frac{p(Y=0)}{p(Y=1)}\right]}$$

Or equivalently:

$$p(Y = 1|X) = \frac{1}{1 + \exp(\sum_{i=1}^n w_i X_i + w_0)}$$

$$\text{where } w_i = \frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2}, \quad w_0 = \ln \frac{1 - P(Y=1)}{P(Y=1)} + \sum_{i=1}^n \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2}$$



GNB Parametric Form of $P(Y|X)$

$$I, II \Rightarrow p(Y = 1|X) = \frac{1}{1 + \exp\left[\sum_{i=1}^n \left(\frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2} X_i + \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2}\right) + \ln \frac{p(Y=0)}{p(Y=1)}\right]}$$

Or equivalently:

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$$p(Y = 0|X) = 1 - p(Y = 1|X) = \frac{\exp(\sum_{i=1}^n w_i X_i + w_0)}{1 + \exp(\sum_{i=1}^n w_i X_i + w_0)}$$



GNB Parametric Form of $P(Y|X)$

If $p(Y = 1 | \mathbf{X}) \geq p(Y = 0 | \mathbf{X})$ Then Class is +

If $\frac{p(Y=1|\mathbf{X})}{p(Y=0|\mathbf{X})} = \frac{1}{\exp(\sum_{i=1}^n w_i X_i + w_0)} \geq 1$ Then Class is +



GNB Parametric Form of $P(Y|X)$

If $p(Y = 1 | \mathbf{X}) \geq p(Y = 0 | \mathbf{X})$ Then Class is +

If $\frac{p(Y=1|\mathbf{X})}{p(Y=0|\mathbf{X})} = \frac{1}{\exp(\sum_{i=1}^n w_i X_i + w_0)} \geq 1$ Then Class is +

If $\exp(\sum_{i=1}^n w_i X_i + w_0) \leq 1$ Then Class is +

If $\sum_{i=1}^n w_i X_i + w_0 \leq 0$ Then Class is +; Else Class is -



GNB Parametric Form of $P(Y|X)$

If $p(Y = 1|X) \geq p(Y = 0|X)$ Then Class is +

If $\frac{p(Y=1|X)}{p(Y=0|X)} = \frac{1}{\exp(\sum_{i=1}^n w_i X_i + w_0)} \geq 1$ Then Class is +

If $\exp(\sum_{i=1}^n w_i X_i + w_0) \leq 1$ Then Class is +

If $\sum_{i=1}^n w_i X_i + w_0 \leq 0$ Then Class is +; Else Class is -

Therefore, naïve Bayes learning algorithm can be viewed as a linear classifier, under the assumption that variance of each feature is independent of the class (i.e if $\sigma_{ik} = \sigma_i$).



References

- ▶ Victor Lavrenko, Lecture on Gaussian Naïve Bayes, University of Edinburgh
- ▶ Tom Mitchell, Generative and Discriminative Classifiers: Naïve Bayes and Logistic Regression, Machine Learning (2nd ed.), Chapter 3, McGraw Hill, 2015